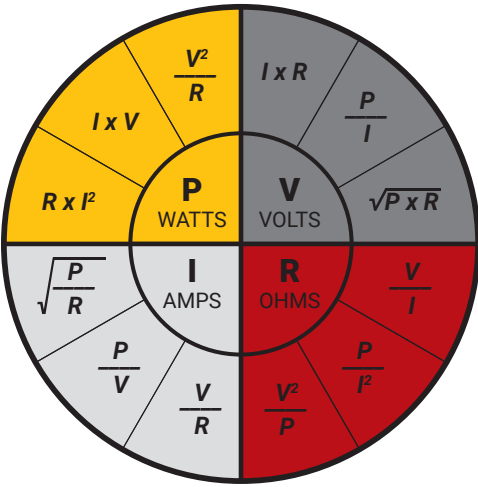


FORMULAS, LAWS, AND EQUATIONS

Ohm's Law



Parallel and Series Formulas

	Capacitor	Resistor	Inductor
Series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$	$R = R_1 + R_2$	$L = L_1 + L_2$
Parallel	$C = C_1 + C_2$	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$	$\frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2}$
Fundamental Formula	$\Delta V = \frac{Q}{C}$	$\Delta V = IR$	$E_L = -L \frac{dI}{dt}$

Voltage Divider Rule

DC Circuits in Series – R is Resistance of load

$$V_i = V_T \frac{R_i}{R_1 + R_2 + R_3 + \dots}$$

AC Circuits in Series – Z is Impedance of load

$$V_i = V_T \frac{Z_i}{Z_1 + Z_2 + Z_3 + \dots}$$

Current Divider Rule for Resistance in Parallel

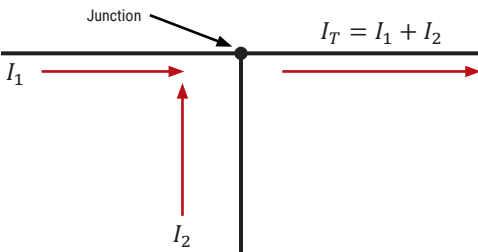
$$I_X = \frac{R_T}{R_X + R_T} I_T$$

Where I_T is the total current entering the node and R_T is calculated as follows:

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

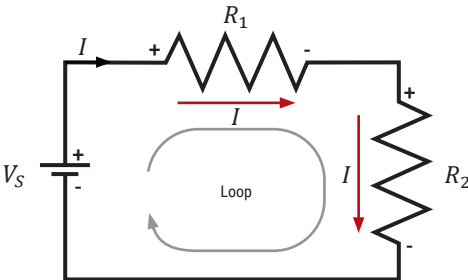
Kirchhoff's Current Law

Current entering a node must equal current exiting that node.



Kirchhoff's Voltage Law

The algebraic sum of all the voltages around any closed loop in a circuit is equal to zero.



$$V_S = IR_1 + IR_2$$

FORMULAS, LAWS, AND EQUATIONS CONTINUED

Coulomb's Law

Electric force F in newtons (N) between two electric charges q_1 and q_2 in coulombs (C) with a distance of r in meters (m).

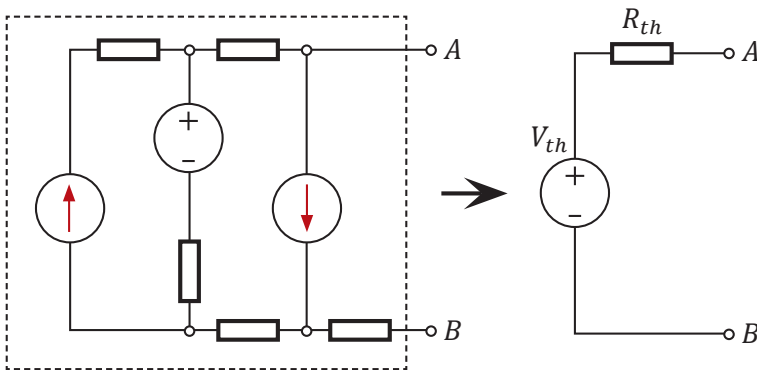
$$F = k \frac{q_1 \cdot q_2}{r^2}$$

Where k is Coulomb's Constant:

$$k = 8.988 \times 10^9 \frac{N \cdot m^2}{C^2}$$

Thevenin's Theorem

Any number of sources or components, no matter how they are connected, can be represented by an equivalent series circuit with respect to any pair of terminals in the network.



Resistance Required for Limiting Current in LED Circuits

Driving LEDs requires a resistor in series from a voltage source to help limit current flow. When you select a resistor for this purpose, choose a device with a power rating between 2 and 10 times the value calculated below in order to avoid excessive resistor temperatures.

$$R = \frac{V_S - V_f}{I}$$

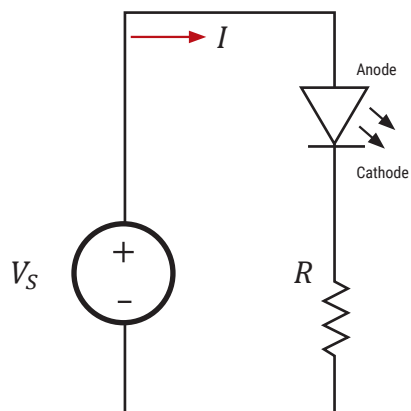
V_S = Supply Voltage Being Used

V_f = Forward Voltage Drop on LED

I = Your Desired Current for LED

(not to exceed max. current rating
and/or power dissipation –
see datasheet for chosen LED)

R = Resistance Needed for Desired Current



Digi-Key offers conversion calculator tools to help you calculate the resistance needed for your design. Find them online at [DIGIKEY.COM/CALCULATORS](https://www.digikey.com/calculators).